Inocetia Sasa

40/40

100%

Module #	671101000-KM-06:
NQF Level	level 4,
Notional hours	130
Credit(s)	Cr 13
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

ROTATING ELECTRICAL MACHINERY
SUMMATIVE ASSESMENT

MI

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Welcome to your Summative Assessment!

This document deals with the summative assessment of your competence towards the outcomes which you need to achieve. "Summative Assessment is assessment for making a judgement about achievement. This is carried out when a learner is ready to be assessed at the end of a programme of learning" (SAQA: Criteria and Guidelines for Assessment Policy Document,).

Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment Sasa

Learner: Moshibudi Inocetia	Manager:
Trainer:	Mentor / Coach:
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

Learners Name: Inocetia Moshibudi Sasa

Learner's signature: Sasa Date: _____

Assessors Name: _____

Assessor's signature: _____ Date: _____

M.I

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____

MI

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: Mashibadi Ivoette Sasa	Signature
Assessors Name:	Signature:

MI

LEARNING UNIT 1:

KM-06-KT01: Rotating electrical machinery - AC motors (35%)

Learning Outcome 1:	KT0101 Construction of alternating current motors
Learning Outcome 2:	KT0102 Principle of operation of alternating current motors
Learning Outcome 3:	KT0103 Configuration of motor connections
Learning Outcome 4:	KT0104 Types of single phase and three phase alternating current motors
Learning Outcome 5:	KT0105 Application of alternating current motors
Learning Outcome 6:	KT0106 Testing principles of single phase and three phase alternating current motors
Learning Outcome 7:	KT0107 Protection of motors
Learning Outcome 8:	KT0108 Calculation of motor properties

Task	Questions Description	Marks
1	a) List the different types of alternating current motors and describe their construction b) Describe with the aid of diagrams, the configuration of motor connections c) Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits.	5 5 10

a) Types of alternating current motors
* Single-phase motors
* three-phase motors

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Construction:
* Stator: Stationery part with windings that produce a magnetic field
* Rotor: rotating part inside stator
* Air gap: space between stator and rotor
* Shaft and bearings: Allow rotation

b) Star (Y) connection
* Windings joined at a common neutral point

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LEARNING UNIT 2:

KM-06-KT02: Rotating Electrical Machinery - DC motors (35%)

Learning Outcome 1:	KT0201	Construction of direct current motors
Learning Outcome 2:	KT0202	Principles of operation for direct current motors
Learning Outcome 3:	KT0203	Configuration of direct current motor connections
Learning Outcome 4:	KT0204	Types and application of direct current motors
Learning Outcome 5:	KT0205	Testing principles of direct current motors
Learning Outcome 6:	KT0206	Protection of direct current motors
Learning Outcome 7:	KT0207	Calculation of direct current motor properties

Task	Questions Description	Marks
2	a) Explain the principle of operation of direct current motors	2
	b) Describe and explain how various tests are conducted on direct current motors.	2
	c) Describe and explain protection devices used for direct current motors	3
	d) Compare the advantages and disadvantages of different types of direct current motors	3

a) A direct current (DC) Motor works on the principle that when a current-carrying conductor is placed in a Magnetic field, it experiences a force. this Force causes the armature to rotate in the direction of rotation is given by Fleming's left hand rule

b) common tests include

* No-load test: motor runs without load to check the speed

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LEARNING UNIT 3:

Rotating electrical machinery - Alternators and Generators (30%)

Learning Outcome 1:	KT0301 Construction of alternators and generators
Learning Outcome 2:	KT0302 Principle of operation of alternators and generators
Learning Outcome 3:	KT0303 Configuration of alternator and generator connections
Learning Outcome 4:	KT0304 Types and application of alternators and generators
Learning Outcome 5:	KT0305 Testing principles of alternators and generators
Learning Outcome 6:	KT0306 Protection of alternators and generators
Learning Outcome 7:	KT0307 Calculate properties of alternators and generators

Task	Questions Description	Marks
2	a) List the different types of alternators and generators and describe their construction	2
	b) Explain the fundamental differences between alternators and generators	2
	c) Describe and explain how tests are conducted on alternators and generators	3
	d) Describe with the aid of diagrams, the configuration of alternator and generator connections, power circuits and control circuits.	3

a) Types of Alternators
1) Salient pole alternator
construction:
-> Rotor: large poles with field windings
-> Stator: laminated core with armature windings
2) Non-Salient pole alternator

2

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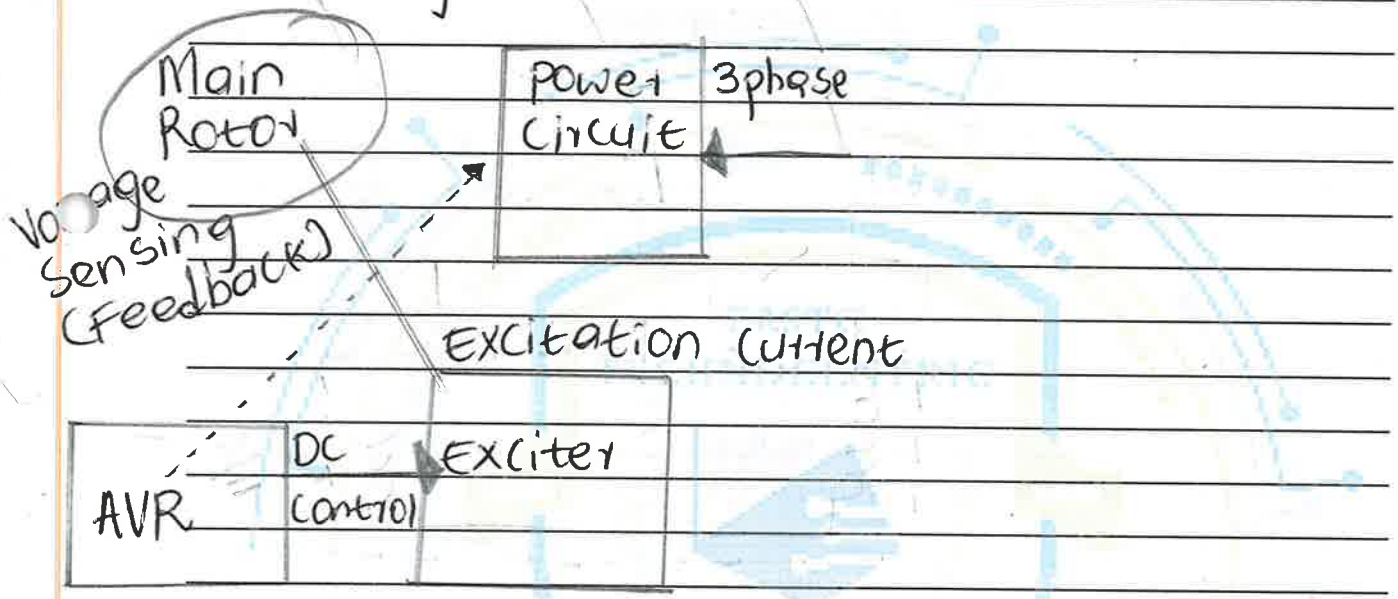
2) Load test

* Apply gradually
* Observe voltage, current and efficiency

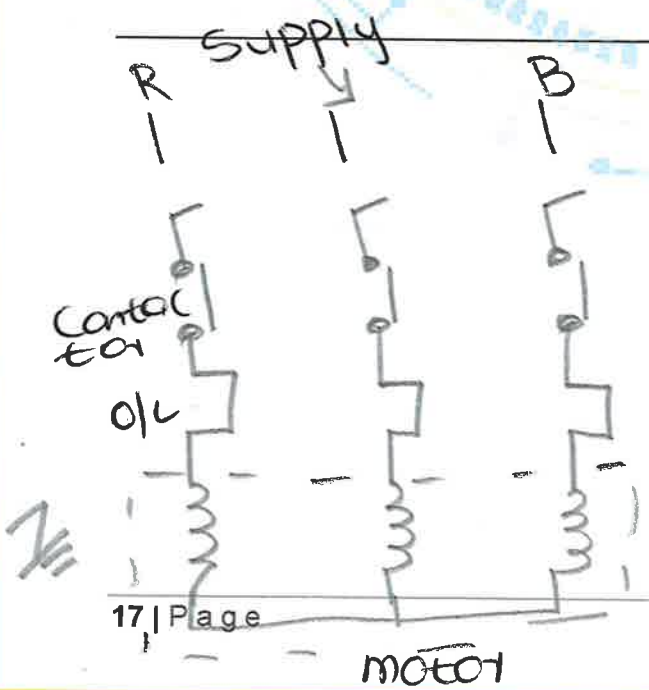
3) Insulation test

* Checks insulation condition using megger

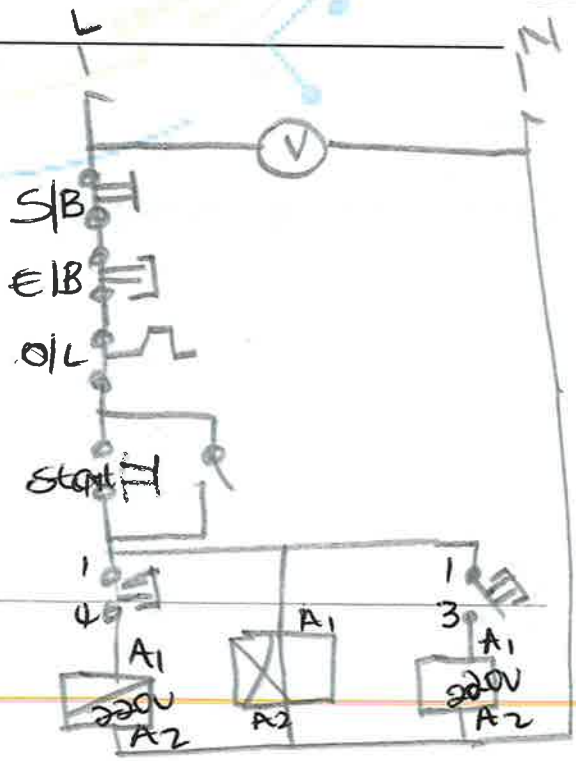
d) Alternator connection diagram



AC Power circuit



DC control circuit



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Candidate Feedback Report

Candidate's Name	Mashibudi Thobetsa Sasa	ID No.	0311110617 082
Assessor's Name		Reg. No.	

ASSESSMENT DECISION

Source of Evidence	C	NYC	Comment
Knowledge questions	✓		good
Product /Work sample	✓		good
Project	✓		good
Indirect Evidence	✓		good
Overall Assessment Decision			
Additional Notes			
Date			
Signature of Assessor		Signature of Candidate	

MI

Candidate Feedback Report

Candidate's Name	Mashibudi Violetta Sasa	ID No.	0311110617 082
Assessor's Name		Reg. No.	

ASSESSMENT DECISION			
Source of Evidence	C	NYC	Comment
Knowledge questions	✓		good
Product /Work sample	✓		good
Project	✓		good
Indirect Evidence	✓		good
Overall Assessment Decision			
Additional Notes			
Date			
			Sasa MP
Signature of Assessor			Signature of Candidate

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Moderator's Report

Moderator's Name		Reg. No.	
Assessor's Name		Reg. No.	
Candidate's Name	Moshiyadi Inocetia Sasa	ID No.	031111061702

MODERATION DECISION

	C	NYC	Comments

Overall Moderation Decision	
Feedback to Assessor	
Action Required	
Date of Moderation	
Signature of Moderator	
Signature of Assessor	
Signature of Candidate	

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